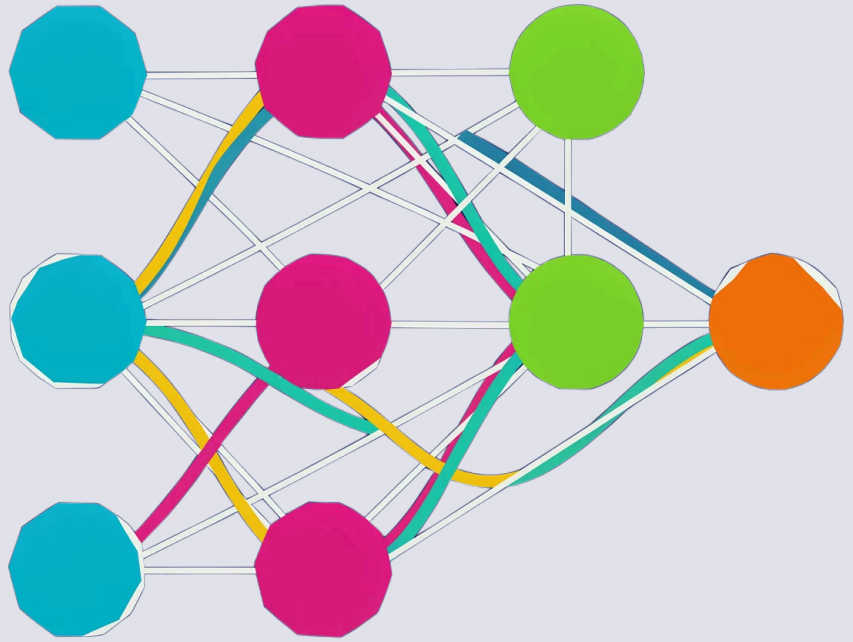
Controlling AI Creativity

Temperature, Top-p & Top-k

Introduction to Generative AI





How an LLM Picks the Next Word

- **The model does not simply pick the one best word every time**
- **It assigns a probability to every possible next word**
- **Then it samples from those probabilities**

Three parameters control this sampling:

The Three Parameters

Parameter	What it controls
Temperature	How "sharp" or "flat" the probability distribution is
Top-p (Nucleus Sampling)	Which portion of likely words is even considered
Top-k	How many of the most probable words are kept in the pool

Case Study: The Ice Cream Shop

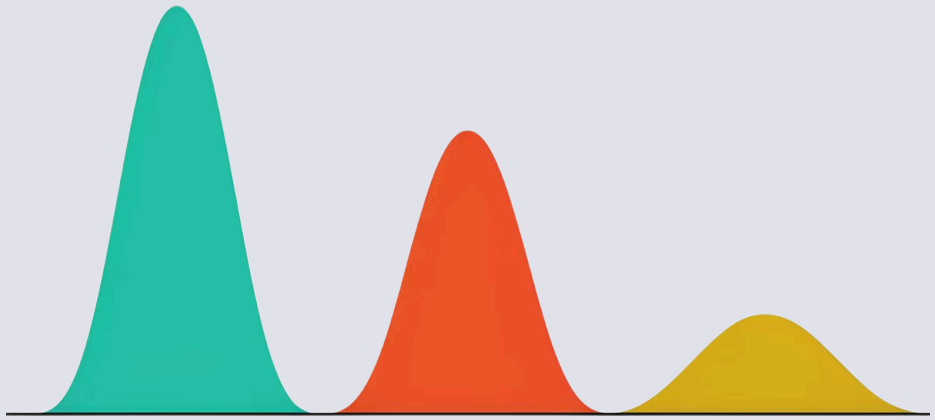
Prompt:

"The chef's favourite dessert is ___"

The model scores many possible next words. Before any sampling:

Word	Probability
chocolate	40%
vanilla	30%
strawberry	15%
pizza	8%
the	4%
elephant	2%
rocket	1%

We'll see how each parameter changes what gets chosen.



Temperature: What It Does

Temperature **rescales** the probability distribution.

Low (0.1 – 0.5)

Sharpens the curve → almost always the top word → **predictable**

1.0

Probabilities used as-is → **balanced**

High (1.5 – 2.0)

Flattens the curve → rare words get a real chance → **creative, unpredictable**

Temperature Analogy: A Bowl of Marbles

Each colour is a word. Blue = chocolate.

Low temperature

The bowl is almost all blue marbles. You nearly always pull blue.

High temperature

The bowl is an even mix of colours. You might pull anything.

Same bowl, same words. Temperature changes how likely each one is.

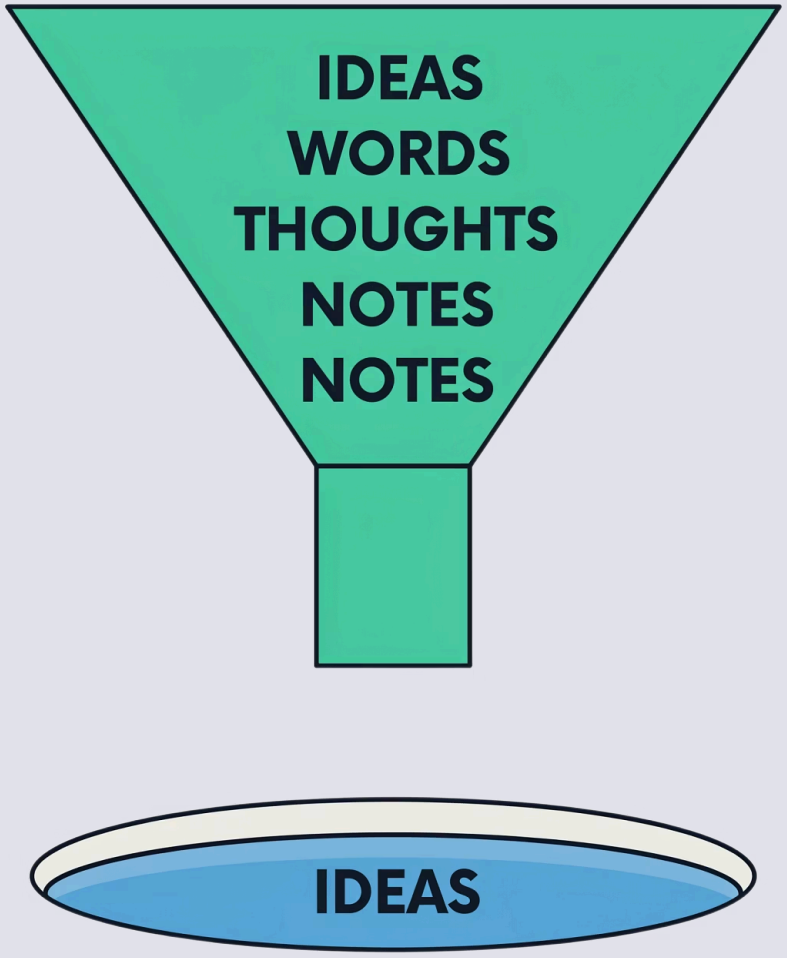


Same Prompt, Different Temperatures

Temperature	Likely output	Character
0.2	"The chef's favourite dessert is chocolate cake."	Safe, expected
1.0	"The chef's favourite dessert is strawberry tart."	Natural, varied
1.8	"The chef's favourite dessert is elephant-shaped pizza."	Weird, creative

Which Temperature Should You Use?

Use case	Recommended temperature
Factual Q&A, coding, maths	0.0 – 0.3
Summarisation, translation	0.3 – 0.7
Creative writing, brainstorming	0.7 – 1.2
Poetry, experimental fiction	1.2 – 2.0



Top-p (Nucleus Sampling): What It Does

Instead of reshaping probabilities, Top-p decides **which words are allowed in**.

01

Sort all words from highest to lowest probability

02

Add up probabilities from the top until the running total reaches p

03

Only words inside that "nucleus" are considered; the rest are discarded

Top-p Example: p = 0.90

Word	Probability	Cumulative	In nucleus?
chocolate	40%	40%	Yes
vanilla	30%	70%	Yes
strawberry	15%	85%	Yes (still below 90%)
pizza	8%	93%	**Yes (first to cross 90%, so we stop here)**
the	4%	97%	No
elephant	2%	99%	No
rocket	1%	100%	No

The model samples from **chocolate, vanilla, strawberry, pizza**, rescaled to sum to 100%.

Top-p Adapts Automatically

The key advantage over a fixed number of words:

Situation	Fixed Top-k = 3	Top-p = 0.90
One word has 95% probability	Still picks from 3 words (wasteful)	Picks only 1 word (focused)
Ten words share probability	Still only 3 (too narrow)	Keeps all 10 (flexible)

Top-p is **context-aware**: the pool grows and shrinks with the model's confidence.

Top-k: What It Does

The simplest of the three:

> "Only consider the **k** most probable words. Ignore the rest."

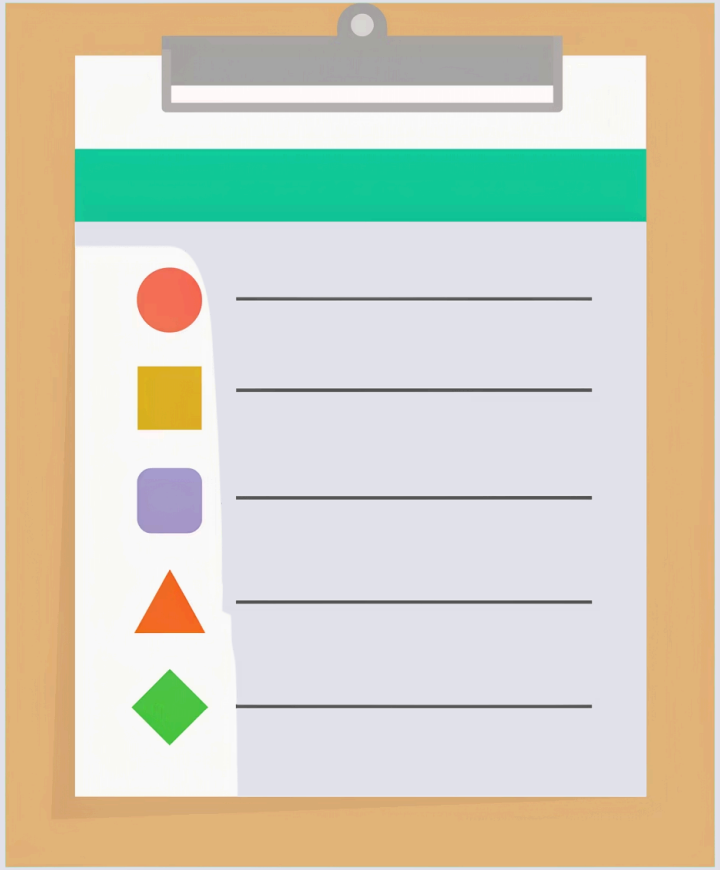
Top-k = 2

chocolate (40%), vanilla (30%)

Top-k = 4

chocolate (40%), vanilla (30%), strawberry (15%), pizza (8%)

A blunt filter compared to Top-p, but useful when you want strict control.



Combining the Parameters

All three can be applied together. A typical order:

01

Top-k: trim the list to k words

02

Top-p: trim further to the nucleus

03

Temperature: reshape the probabilities of the survivors

04

Sample: pick one word



Worked Example: Setup

Prompt: *"Once upon a time, the wizard said: ___"*

Settings:

- temperature = 0.8
- top_p = 0.92
- top_k = 50

Worked Example: Step by Step

1

Top-k

Keep the 50 most probable words ("magic", "hello", "beware", "look", "silence", ...)

2

Top-p

Running sum hits 92% after 12 words, so only **12 words** remain ("magic" 25%, "beware" 18%, "silence" 14%, "come" 10%, ...)

3

Temperature 0.8

Slight sharpening. "magic" rises a bit; very rare words shrink further.

4

Sample

The model picks **"beware"** (second most likely, but never guaranteed)

"Once upon a time, the wizard said: beware"

Quick Summary

Parameter	Range	Low value	High value
Temperature	0.0 – 2.0	Deterministic, safe	Random, creative
Top-p	0.0 – 1.0	Narrow word pool	Wide word pool
Top-k	1 – ∞	Strict (only top words)	Loose (many words allowed)

Analogy Recap



Temperature

Volume knob on a speaker: up for loud and wild, down for quiet and safe



Top-p

A party guest list: invite people whose combined popularity covers 90% of the crowd



Top-k

A shortlist: "I'll only consider the top 5 candidates"

Classroom Experiment

Try this in any playground or chat tool that exposes these settings:

01

Start with temperature = 1.0, top_p = 1.0 (pure sampling, no filtering)

02

Lower temperature and watch the output become predictable

03

Lower top_p to 0.5 and watch it become more focused

Run the same prompt 5 times at each setting. What changes?





Quick Check: Discuss in Pairs

1. What happens to the word "elephant" as temperature goes from 0.2 to 1.8?

2. Why is Top-p better than a fixed Top-k when the model is very confident?

3. Which settings would you pick for a coding assistant? For a poetry generator?

4. In the ice cream example, which words survive top_p = 0.90?